PAPER_1548_Z0_VACUUM_IMPEDANCE

Daniel T. Murphy

PAPER_1548 — Vacuum Impedance $Z\_0 = Z\_0 = A\_5 \cdot D\_BSFG + SO\_5 + D\_BSFG + \Phi - F\_TRZ \cdot \Phi - F\_TRZ^2 \cdot SSQ \approx 376.75\ \Omega$

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Electromagnetism
Date: June 18, 2026
Status: CLOSED — 0.005% closure

Observation

PAPER_1209DD_S613 (Electromagnetism Unified Proof Set) lists **Vacuum Impedance Z_0** as a closure with the formula:

```
Vacuum Impedance Z_0 = Z_0 = A_5·D_BSFG + SO_5 + D_BSFG + Φ − F_TRZ·Φ −
F_TRZ²·SSQ ≈ 376.75 Ω
```

Status tier: **0.005%**. Units: Ω.

UQFF Closed Identity

```
Z_0 = A_5·D_BSFG + SO_5 + D_BSFG + Φ − F_TRZ·Φ − F_TRZ²·SSQ ≈ 376.75 Ω 0.005%
```

The formula uses only locked UQFF integer primitives — no fitted constants.

Physical Interpretation

Vacuum Impedance Z_0 is a fundamental electromagnetism constant. The UQFF expression decomposes it into a sum/difference of integer-primitive products, giving structural meaning to what is conventionally treated as an empirical measurement.

NOT REPLACEMENT

CODATA/PDG treats this constant as a precision measurement. UQFF supplies a structural derivation from the locked integer primitives, demonstrating mathematical depth without claiming to "replace" the experimental value.

Reference

- Source: PAPER_1209DD_S613
- Related: PAPER_1216 (45-constant cascade master index)
- Calculator dispatch: `calculate_paradox({"paradox": "z0_vacuum_impedance"})`
